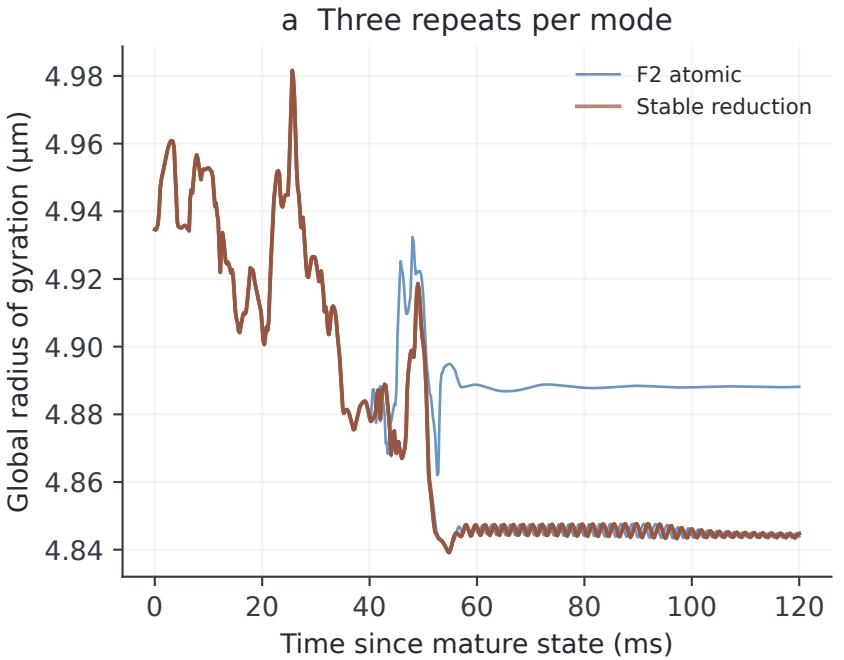
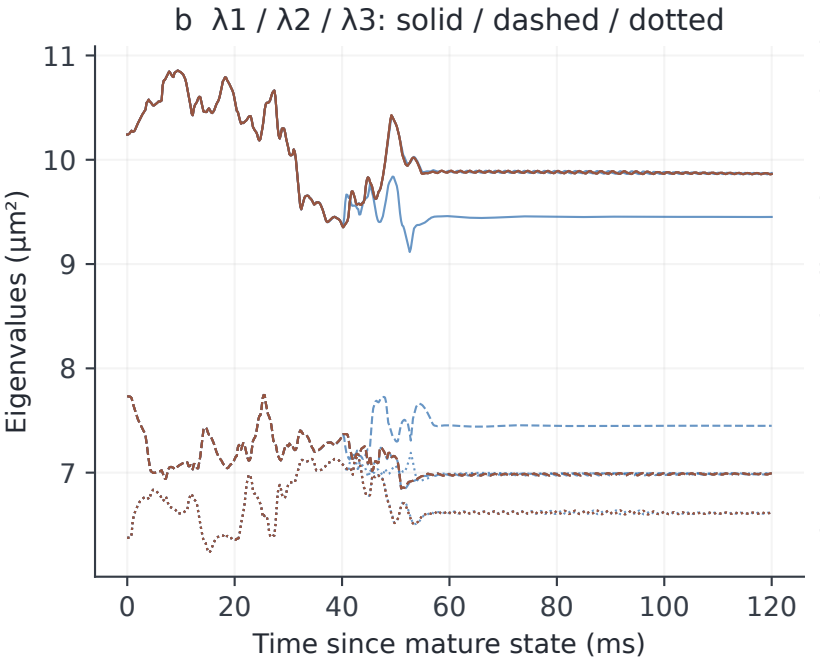
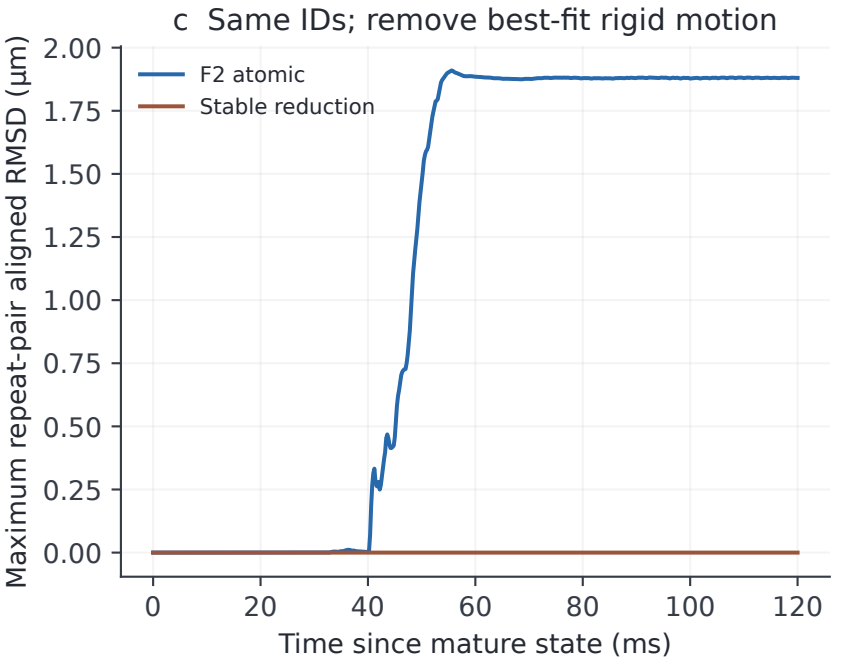
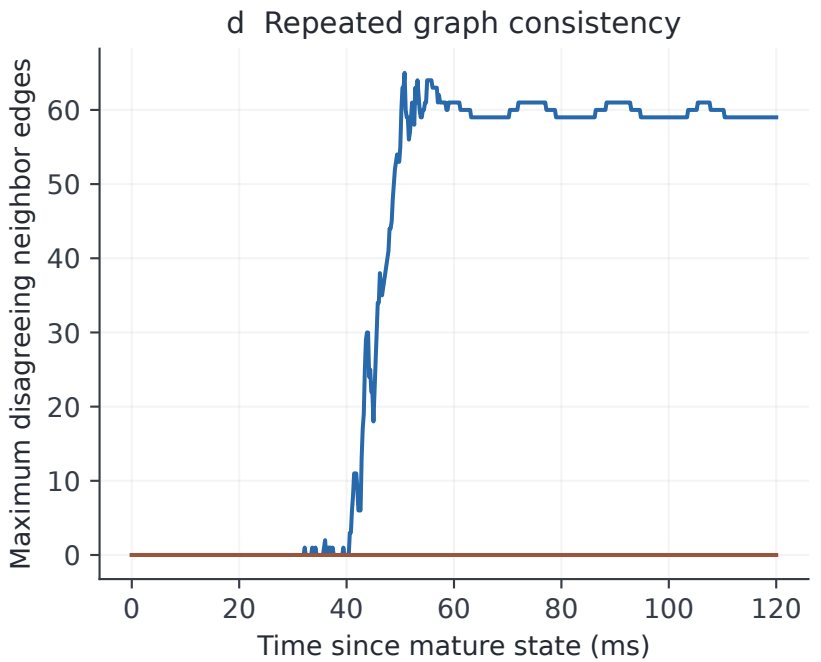
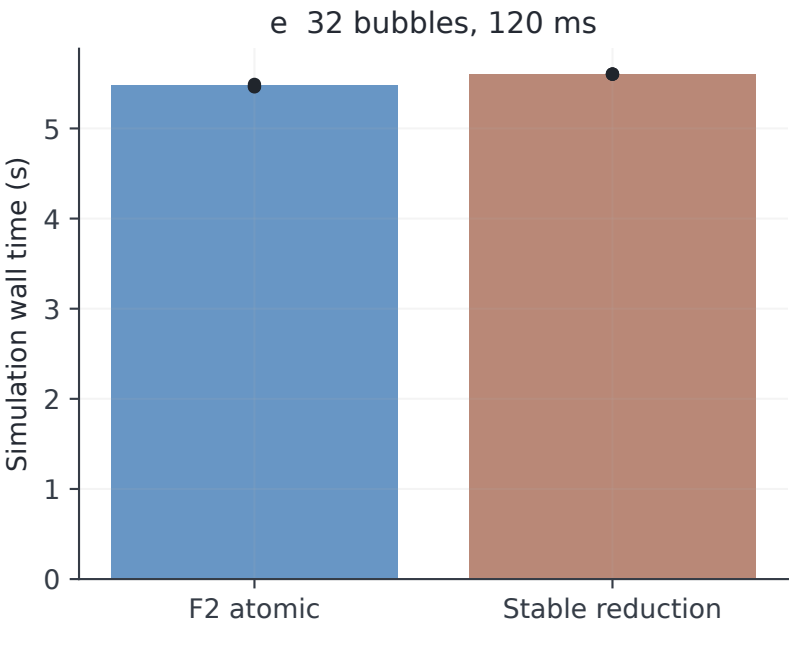
F3-2 | Numerical reproducibility with frozen physics
R = 1.78 μm · Δt = 10 μs · same saved 32-bubble state · same GPU











f What the checks resolve

Eight fixed-state repetitions:
Atomic pair force varies by ≤7.76 × 10^{-26} N.
Stable pair force is bitwise identical.
Component-only stabilization retains variation.

Three full stable trajectories:
All positions and sorted edge events identical.
Graph and direct short replay also agree.

Separate 1 pm / 1 nm initial perturbations
still yield ≈3.42 / 3.43 μm internal differences.
Repeatability does not imply physical accuracy.